

ADEKUNLE ADELAKUN

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EDUCATION

- **Federal University of Agriculture, Abeokuta** *Oct 2019 – Oct 2024*
B.Sc. Computer Science

SKILLS

- **Languages/Tools:** Python, SQL, Docker, FastAPI, MLflow, Pandas, NumPy, Matplotlib
- **Frameworks:** TensorFlow, PyTorch, LangChain, LlamaIndex, PydanticAI, CrewAI
- **Other:** AWS (EC2), Git, CI/CD, Streamlit

WORK EXPERIENCE

- **Data Science Nigeria** *Machine Learning Engineer — May 2025 – Present*
 - Generated training data and improved response quality using advanced prompt engineering techniques.
 - Developed and evaluated audio classification models for speech-based datasets to enhance content accessibility.
 - Automated audio duration extraction and preprocessing pipelines, improving data handling efficiency.
- **SingleEye Technologies** *Machine Learning Engineer — January 2025 – April 2025*
 - Contributed to the development of the AI engine powering SingleEye's products.
 - Built and optimized machine learning pipelines for computer vision and natural language processing tasks.
 - Deployed scalable ML models to production with Docker and FastAPI.

PROJECTS

- **Loan Approval Prediction Deployment** *FastAPI, Docker, AWS, ML*
 - Built ML model to predict loan approval outcomes using applicant financial and demographic data.
 - Created a REST API with FastAPI to serve predictions for seamless integration with external apps.
 - Containerized with Docker and deployed on AWS EC2, ensuring scalability and reliability.
 - Applied preprocessing, feature engineering, and evaluation to boost accuracy and robustness.
 - Reduced manual application review time by ~40% and minimized default risk for lenders.
- **Conversational RAG Chatbot with PDF Support** *LangChain, FastAPI, Docker, ChromaDB*
 - Developed Retrieval-Augmented Generation (RAG) API for intelligent Q&A over PDFs.
 - Implemented document loading, text chunking, and vector storage (ChromaDB) for efficient retrieval.
 - Deployed with FastAPI and Docker for production-ready scalability and reliability.
 - Enabled multi-session queries with context-aware responses, improving usability.
 - Reduced manual PDF review time by ~60%, boosting decision-making efficiency.
- **Facial Emotion Detection System** *Vision Transformer, FastAPI, Streamlit*
 - Developed deep learning system to classify emotions from facial images.
 - Leveraged vision transformer for feature extraction and classification.
 - Fine-tuned model with data augmentation, learning rate scheduling, and dropout.
 - Deployed using FastAPI and Streamlit for a web-based interface.
- **AI-Powered Email Processing System** *FastAPI, SQLAlchemy, Ollama, Streamlit*
 - Built an end-to-end email pipeline that ingests real Gmail emails, persists them, and tracks processing state.
 - Implemented intent classification using a local LLM (Ollama) with structured JSON output and confidence scores.
 - Generated draft responses based on classified intent and stored AI decisions for traceability.
 - Exposed REST APIs with FastAPI and Pydantic schemas, including email retrieval and filtering via Swagger UI.

LEADERSHIP

- **Deep Learning Indaba 2024**

Fundraising Volunteer

- Contributed to organizing and fundraising efforts for Africa's premier ML/AI community event.

PROFESSIONAL ACTIVITIES

- Open Source Contributor
- AI Community Engagement